Pharmaceutical Analytical Techniques

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Pharmaceutical Analytical Techniques (A33681)

Short Title:	Pharm Anal Tech		
Faculty	Science and Computing		
Department	Science		
Cluster	Pharmaceutical Science		
Author	AHAYDEN		
Credits	10	**Level:**	Advanced

Description of Module / Aims

This module will encompass the area In-process sampling and testing and the use of various analytical instrumentation for off-line and on-line testing. The learner will get an insight into critical processing parameters and the modern PAT techniques utilised to control Critical Quality Attributes in Pharmaceutical and Biopharmaceutical processing. The regulation around PAT and various PAT case studies will be evaluated and reviewed.

Programmes

PHAR-0023	BSc (Hons) in Good Manufacturing Practice and Technology (WD_SGMPT_B)	4 / 1 / M

Indicative Content

- Overview of the type and importance of In-Process sampling and testing for Quality products
- Application of analytical techniques in the analysis of In-Process samples.
- Use of modern Process Analytical Technologies for Pharmaceutical and Biopharmaceutical Processes.
- Regulations, Risk Management and Case Studies of on-line PAT applications.
- Operation of PAT laboratory instrumentation for sample analysis.
- Use of UV, MIR, NIR, RAMAN, and ATR, spectroscopic techniques.
- Particle sizing, rheology and use of sensors for process monitoring and online analysis.
- Introduction to Data acquisition and Chemometrics for PAT

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise and comment on the critical nature of IPC sampling and testing and real time analysis as well as the regulations around PAT.
2. Operate and use some practical PAT techniques to evaluate product quality.
3. Evaluate and discuss different case studies in the application of PAT.
4. Appraise and contrast the different PAT techniques employed in the Pharmaceutical and Biopharmaceutical Industry.
5. Interpret and analyse analytical spectra and data.

Learning and Teaching Methods

- Lectures
- Laboratory-based experiments

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5
Continuous Assessment	50%	
Assignment	20%	1,2,3
Lab Report	30%	4,5

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of IPC sampling and Process Analytical Technology. Lack of competence with regard to associated laboratory skills, practices and instrumentation.

40%-49%: Will be able to show a limited understanding of the fundamentals of Process Analytical Technology, but may have some difficulty with applying this theory. Will display some ability with regard to associated laboratory skills, practices and instrumentation.

50%-59%: As well as the above, the student will be to show a reasonable understanding of the complexities of PAT and real time analysis. Will display reasonable competence with regard to associated laboratory skills, practices and instrumentation.

60%-69%: In addition, the student will demonstrate a more detailed knowledge of the PAT, and the associated equipment and regulatory guidelines. Will display very good competence with regard to associated laboratory skills, practices and instrumentation.

70%-100%: All of the above to an excellent level. In addition, the learner will be able to demonstrate comprehensive knowledge and understanding of IPC sampling and testing, PAT and regulatory guidelines as well as the PAT instrumentation that is used. Will be able to critically evaluate case studies in the area. Will display advanced competence and independence with regard to associated laboratory skills, practices and instrumentation.

Note: In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Lab		12
Independent Learning		234

Essential Material

- Bakeev, K.A. _Process Analytical Technology: Spectroscopic Tools and Implementation Strategies for the Chemical and Pharmaceutical Industries._. 2nd. UK: Wiley, 2010.